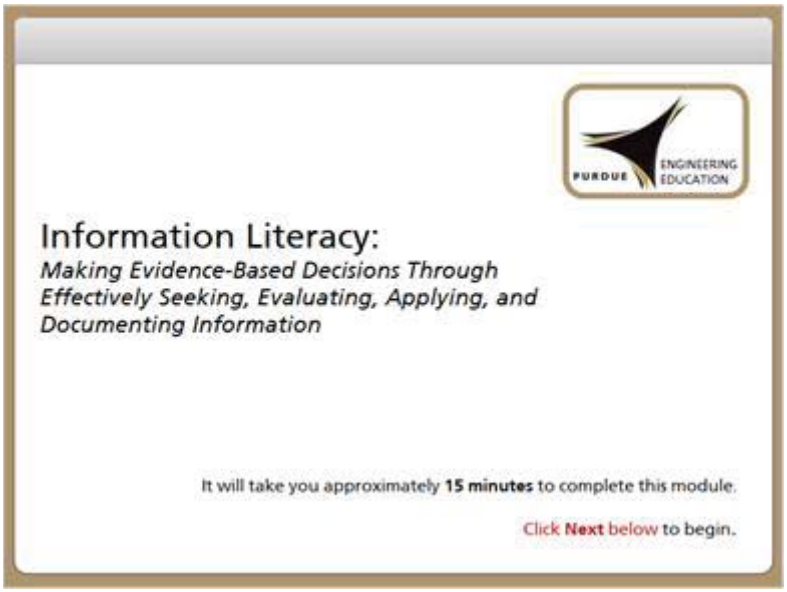
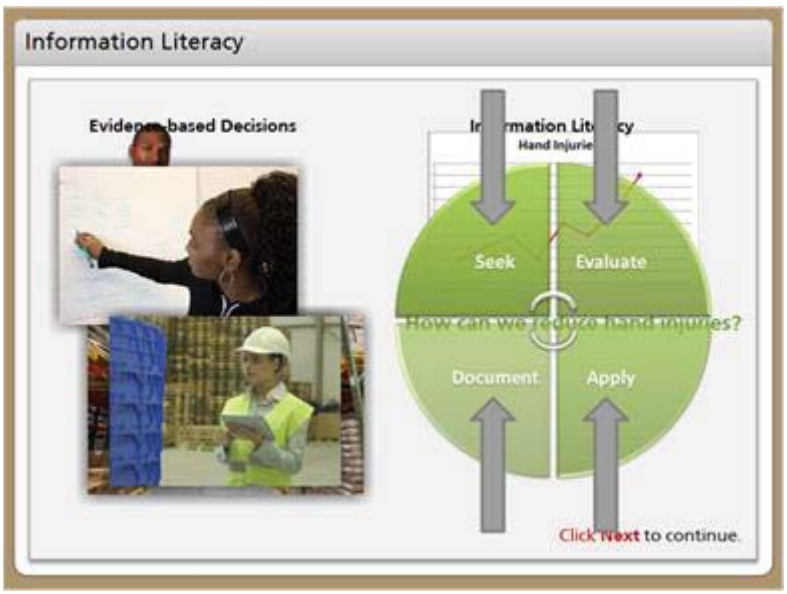
Information Literacy

1. Introduction

1.1 Welcome



1.2 Information Literacy



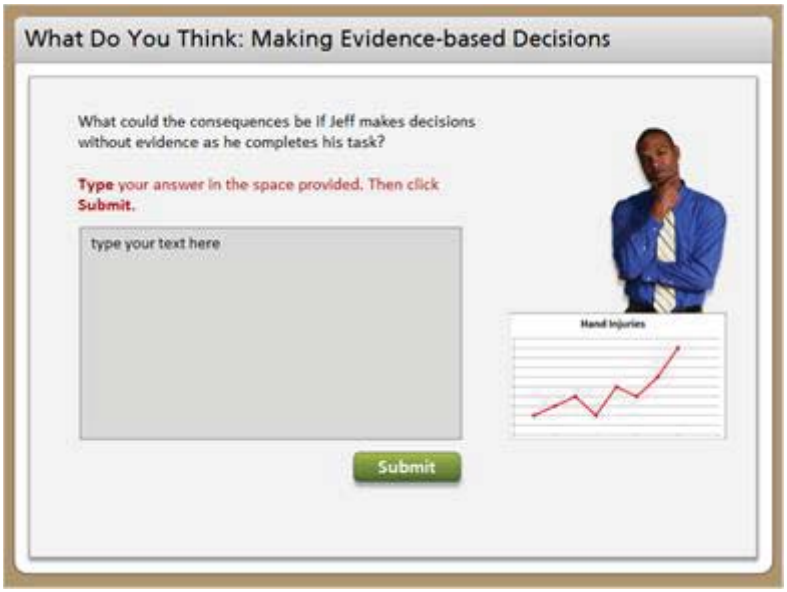
Notes:

Jeff is a Purdue industrial engineering student who has a summer job as an engineering intern in a facility that manufactures and packages foods. In the past year, the facility has seen an increase in the number of hand injuries to workers in the shipping and receiving department.

One of Jeff's projects for the summer is to identify ways to reduce the frequency of hand injuries. By the end of the summer, Jeff has been asked by the facility's safety manager to analyze the current situation and identify specific ways to reduce hand injuries.

Both engineering students and practicing engineers make evidence-based decisions. Making evidence-based decisions involves seeking information, evaluating it, applying it to the situation, and documenting it. We refer to the ability to complete these four tasks effectively as "information literacy."

1.3 What Do You Think: Making Evidence-based Decisions



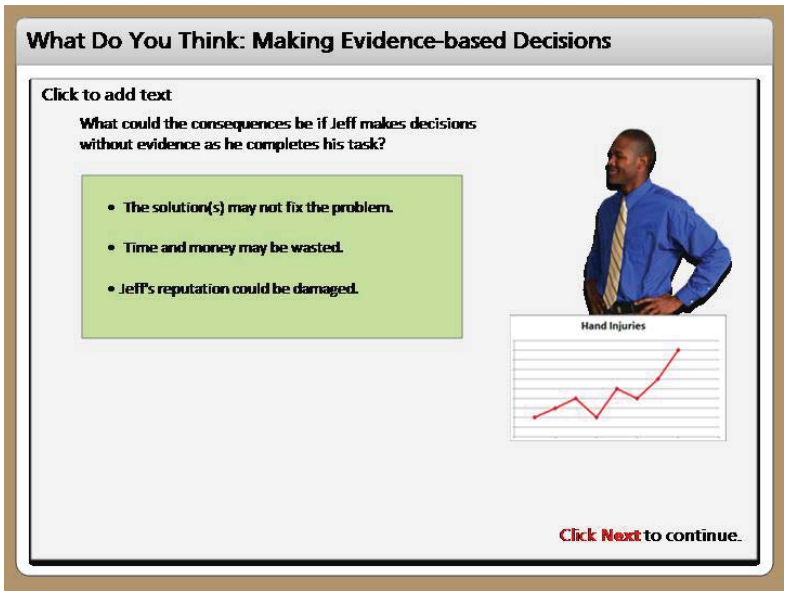
Notes:

Think about Jeff and the project he has been assigned. What could the consequences be if Jeff makes decisions without evidence as he completes his task? Type one or two consequences in the space provided, and then click Submit to compare your answers.

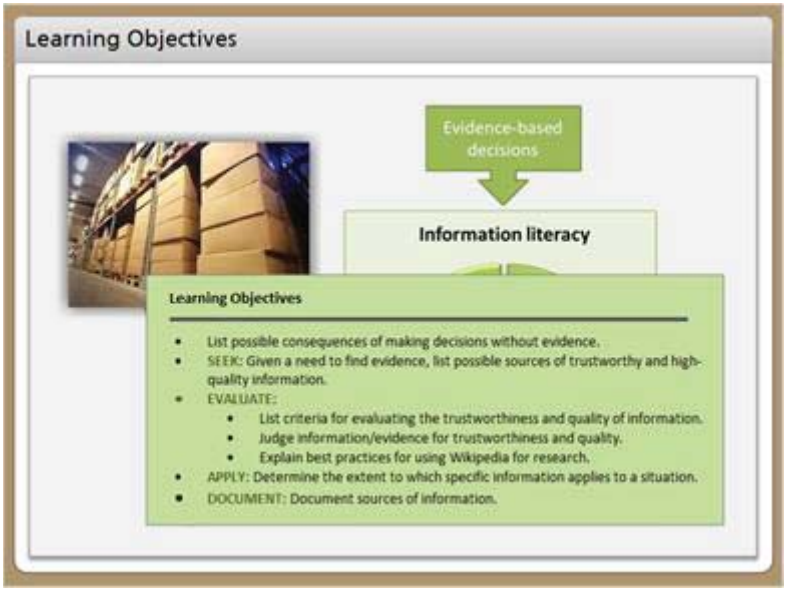
If Jeff completes his task without making evidence-based decisions, negative consequences could occur.

For example, the solution or solutions Jeff proposes or identifies may not fix the problem; they may even make the problem worse and cause more injuries. Time and money may be wasted, and Jeff's reputation as an engineer could be damaged. These are just a few of the consequences that illustrate the importance of making decisions based on evidence.

Submit (Slide Layer)



1.4 Learning Objectives



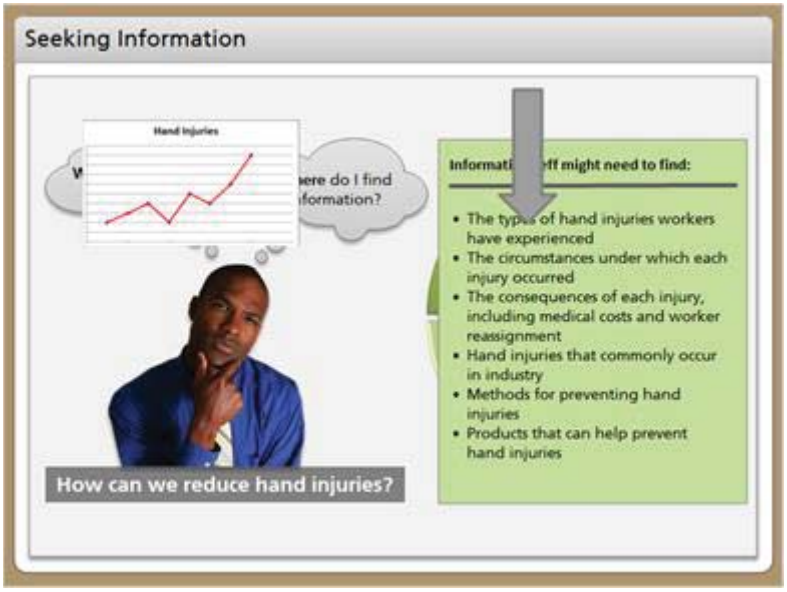
Notes:

As you can see, it is important that Jeff make evidence-based decisions while drawing on his information literacy skills to successfully complete his task. In this module, you will learn more about each information literacy task and how Jeff executes them for his project.

Review the learning objectives shown here, and then click **Next** to begin.

2. SEEK

2.1 Seeking Information



Notes:

It is Monday morning and Jeff is ready to get started with his task of analyzing the current situation related to hand injuries and identify specific ways to reduce these injuries. Jeff will begin by seeking information. Seeking information involves answering the two questions shown here.

Jeff has already thought about the first question and has created a list of some the information he might need to find. Review the list shown here, and then continue to the next page.

2.2 What Do You Think: Information Sources

(Multiple Response, 10 points, 1 attempt permitted)

Correct	Choice
X	Interviews with the plant safety manager and employees who have been injured and their supervisors
X	Existing reports or data the company has about the injuries
X	Government reports on hand injuries and how to prevent them, such as from the Occupations Safety and Health Administration (OSHA)
X	Technical reports on hand injuries or article on hand injuries in trade magazines

Feedback when correct:

Correct! All of these information sources could be useful to Jeff. Continue to the next page to learn more about seeking information.

Feedback when incorrect:

Sorry, that isn't correct. All of these information sources could be useful to Jeff. Continue to the next page to learn more about seeking information.

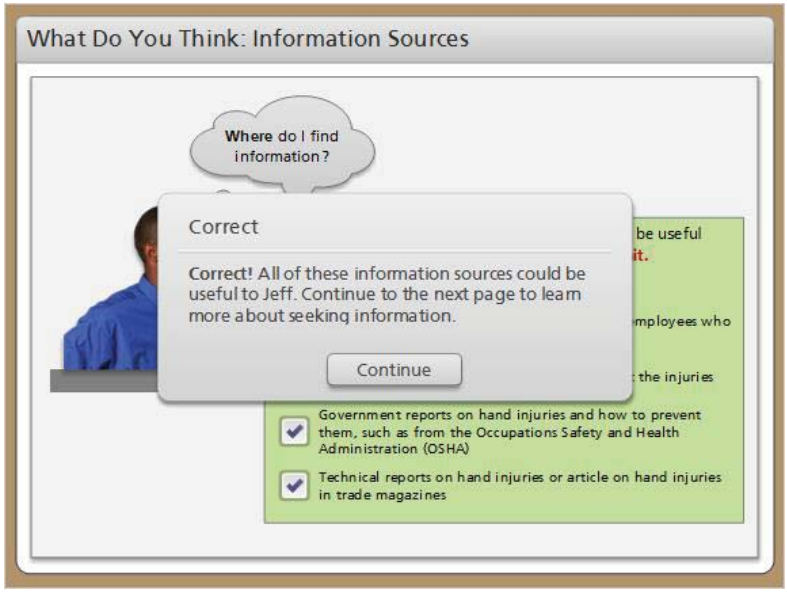
Notes:

Now that Jeff has an idea of the information he needs, he is ready to answer the next question:

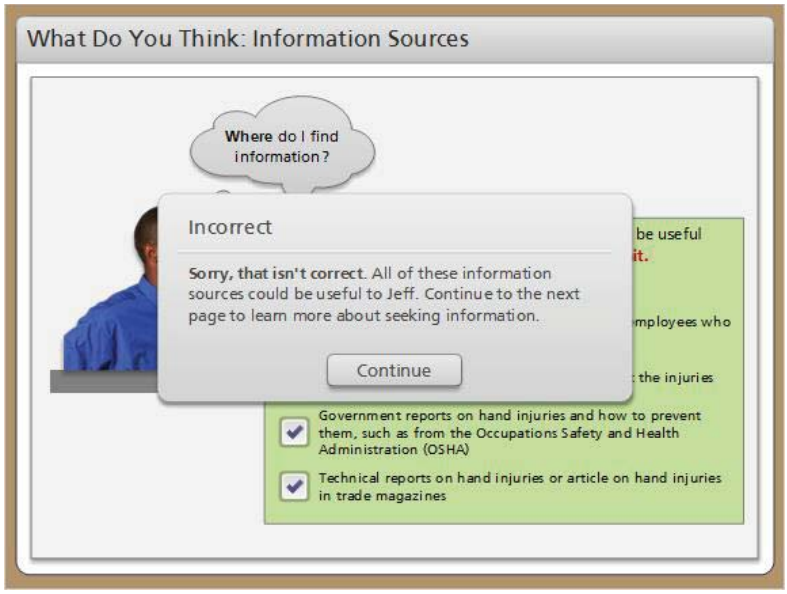
“Where do I find information?”

Which of the following information sources do you think could be useful to Jeff? Select all that apply, and then click Submit.

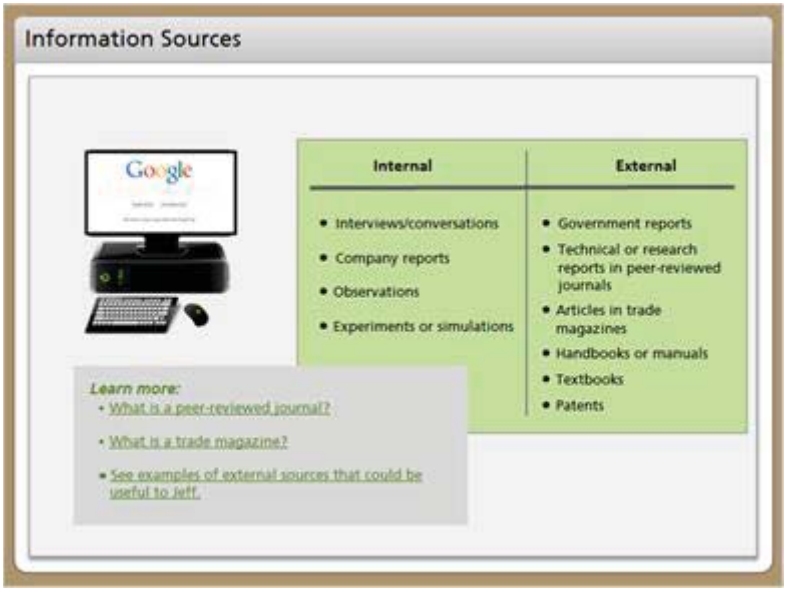
Correct (Slide Layer)



Incorrect (Slide Layer)



2.3 Information Sources



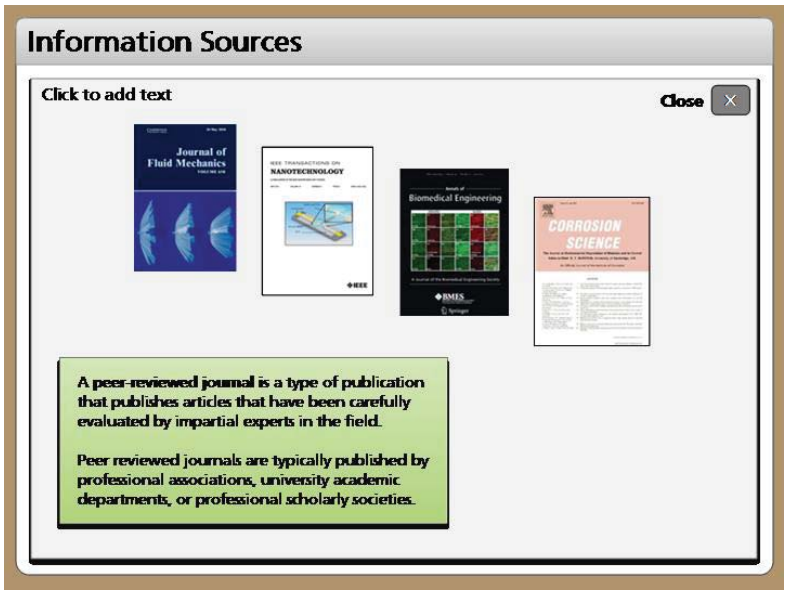
Notes:

When conducting research, your first thought might be to search the internet. This is not a bad idea, because huge amounts of information are readily and conveniently available to us on the Internet. However, it is important to consider other sources of information as well.

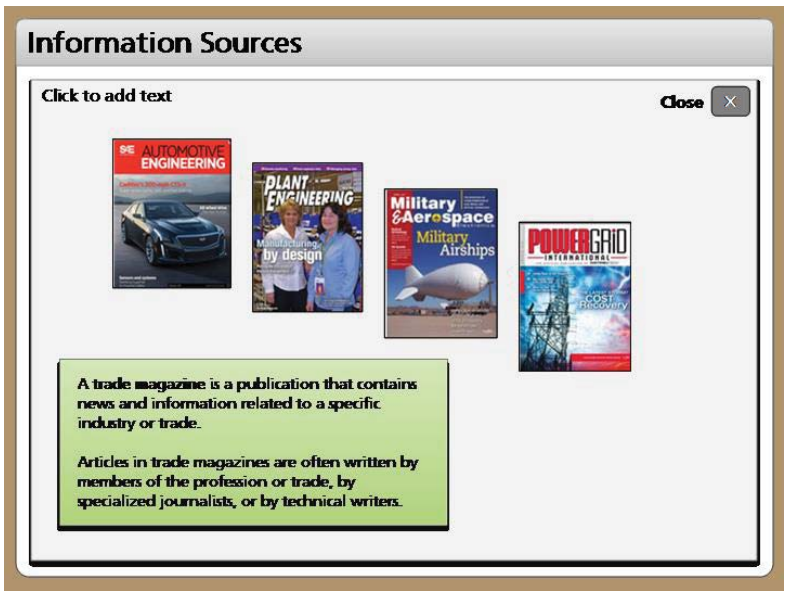
One way to classify information is as either internal to the organization or external to the organization. We just identified several sources of internal information that could be useful to Jeff, including interviews or conversations with workers, supervisors, or the company safety manager or company reports about injuries or work processes. Other internal sources of information include observations, experiments, or simulations that Jeff conducts himself. Notice that some of these internal sources of information, such as interviews or observations, can be informal, but they can still be very important sources of useful information.

We've also already identified some external sources of information, including technical or research reports and articles. Other external sources of information include handbooks or manuals, textbooks, or patents.

Peer Review (Slide Layer)



Trade Magazine (Slide Layer)



2.4 Purdue Libraries

Notes:

While some of the external sources of information we just identified, such as government reports and journal articles, may be available freely on the Internet, some are not.

However, as a Purdue student, you can access through the Purdue Libraries almost 5 million print and electronic books and almost another 5 million articles, reports, and research documents. Many of these items can be easily accessed through the library website.

2.5 Google Scholar

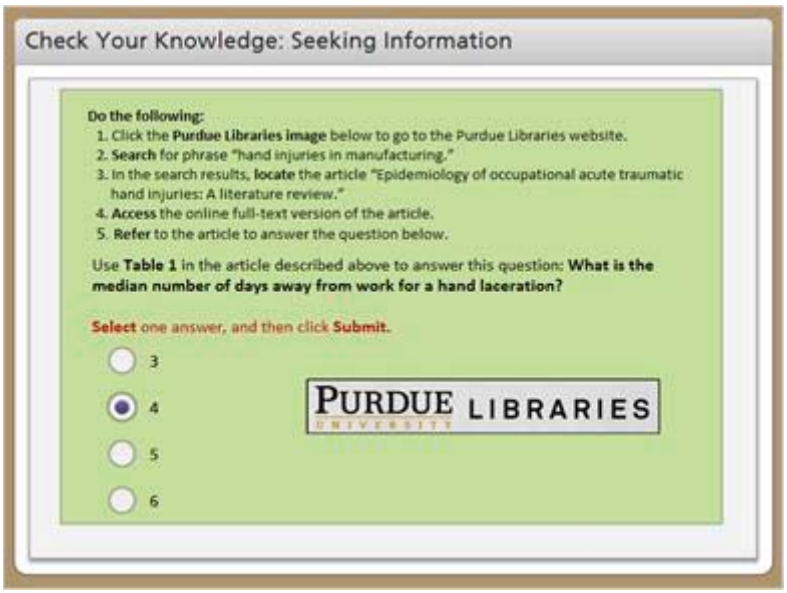
Notes:

Another resource for seeking information is Google Scholar. Google Scholar is a targeted search for scholarly literature, including articles and books from academic publishers.

A nice feature of Google scholar is the ability to set your library preference. For example, if you set your library preference to Purdue, Google Scholar will tell you if a resource is available through the Purdue Libraries. Often, you can click the link in the Google Scholar search results and directly access the resource through the Purdue Libraries.

2.6 Check Your Knowledge: Seeking Information

(Multiple Choice, 10 points, 1 attempt permitted)



Correct	Choice
	3
X	4
	5
	6

Feedback when correct:

That's right! You selected the correct response. As you can see, the Purdue Libraries are a useful and valuable resource to you as a student. Use this resource frequently as you seek information.

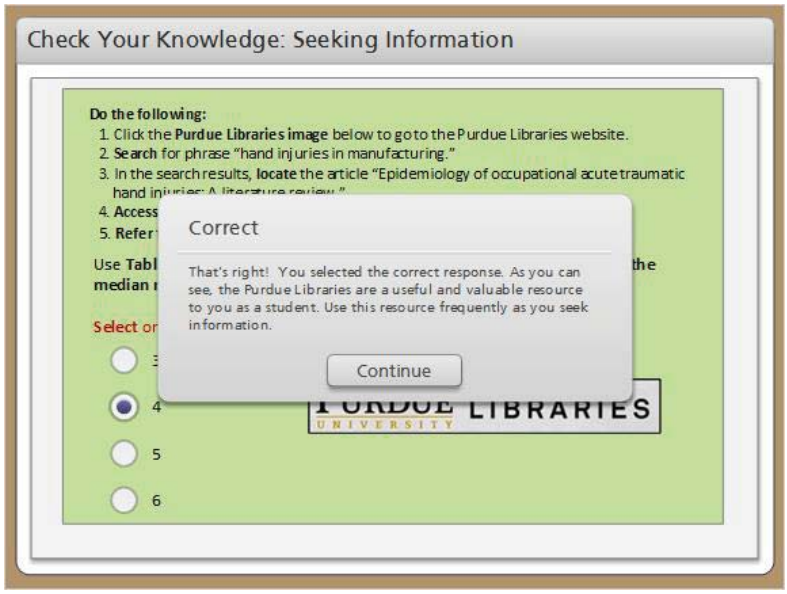
Feedback when incorrect:

You did not select the correct response; you might want to check the article again. As you can see, the Purdue Libraries are a useful and valuable resource to you as a student. Use this resource frequently as you seek information.

Notes:

Practice seeking information using the Purdue Libraries website by answering the question here.

Correct (Slide Layer)



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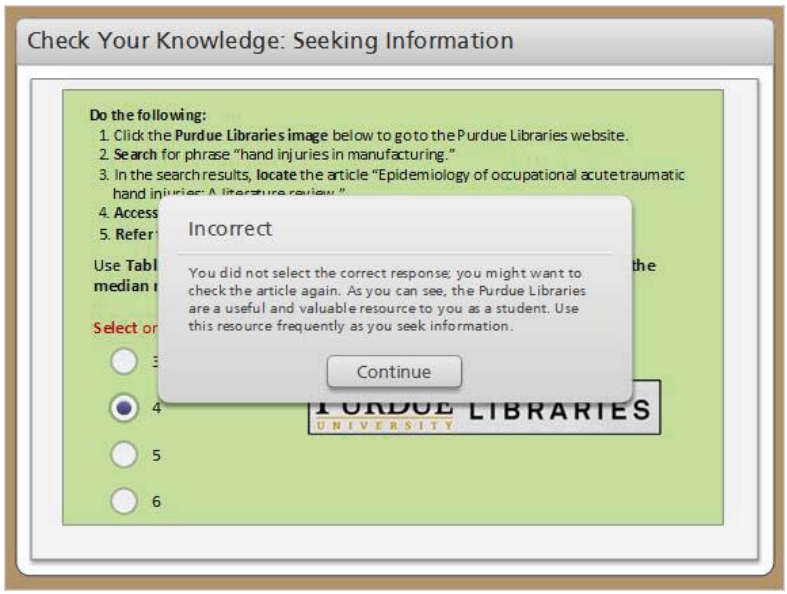
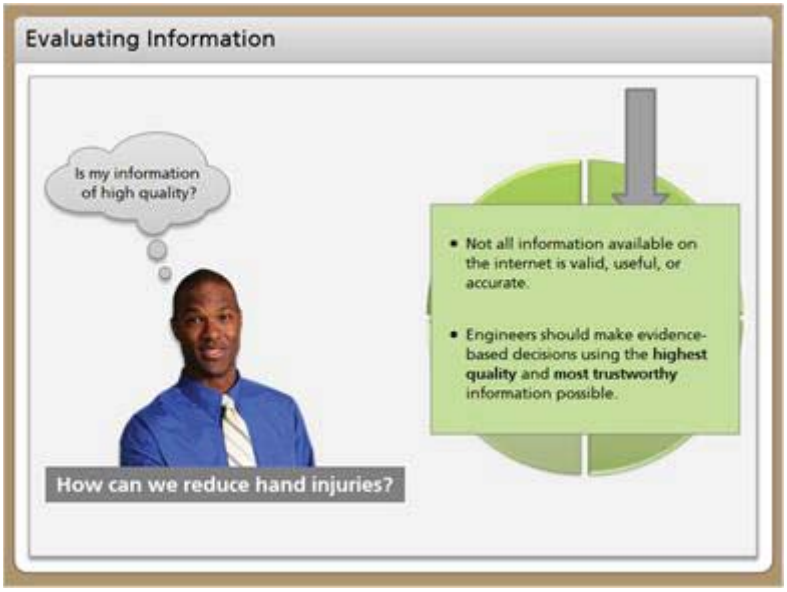


3. EVALUATE

3.1 Evaluating Information



Notes:

As Jeff seeks and gathers information on how to reduce hand injuries, he must continually evaluate the quality of the information. Evaluating information is the second information literacy task.

This tasks is important for several reasons. First, even though the Internet has made huge amounts of information readily available to us, not all of this information is valid, useful, or accurate.

Second, engineers should not just make evidence-based decisions, they should make evidence-based decisions with the highest quality and most trustworthy information possible.

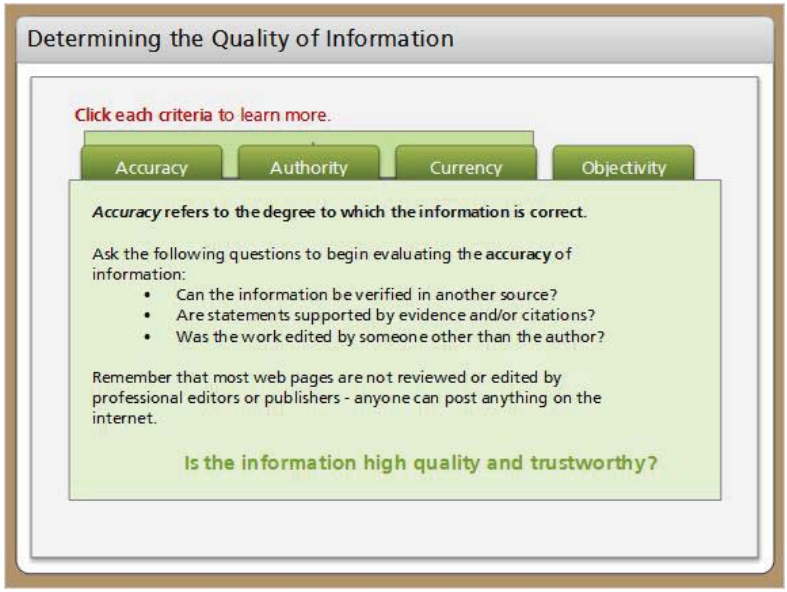
3.2 Determining the Quality of Information

Notes:

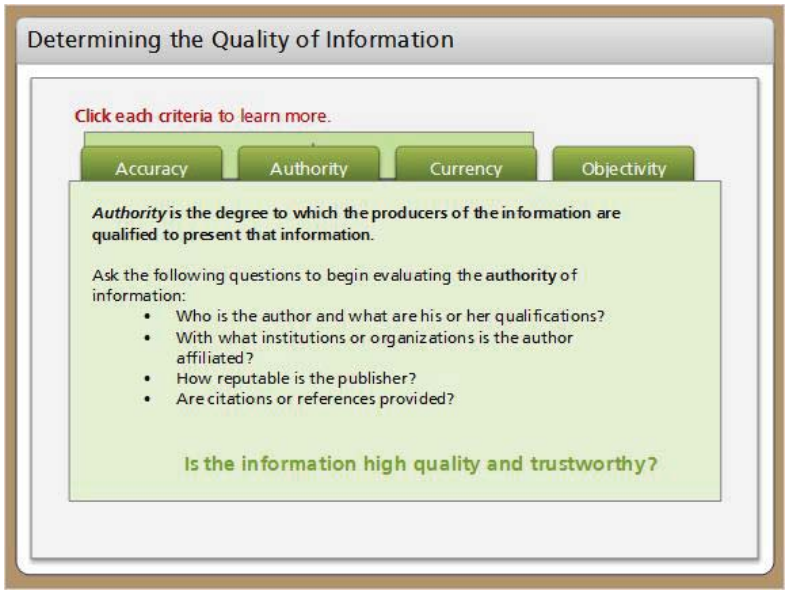
Earlier you learned that information can come from both internal and external sources; some of these sources are listed here. So how can you determine if information is high quality and trustworthy?

The four criteria shown here can help you evaluate information. Click each criteria to learn more.

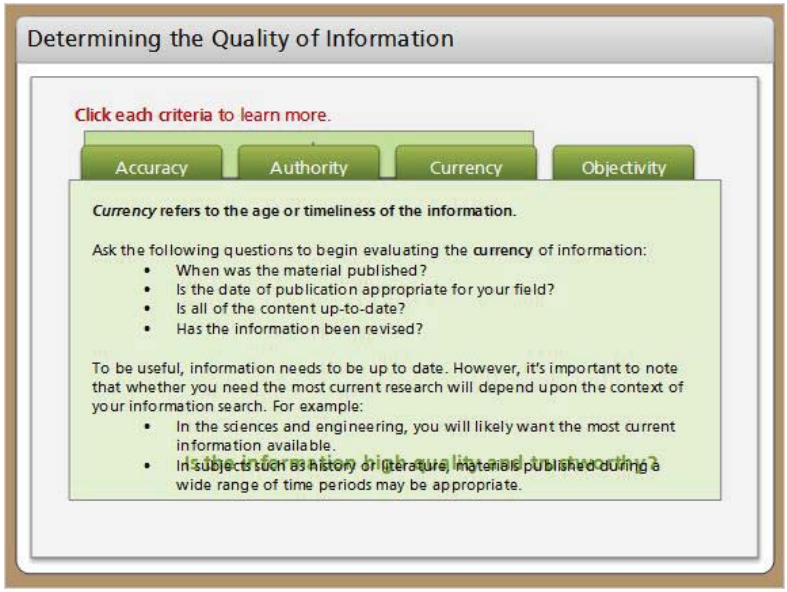
Accuracy (Slide Layer)



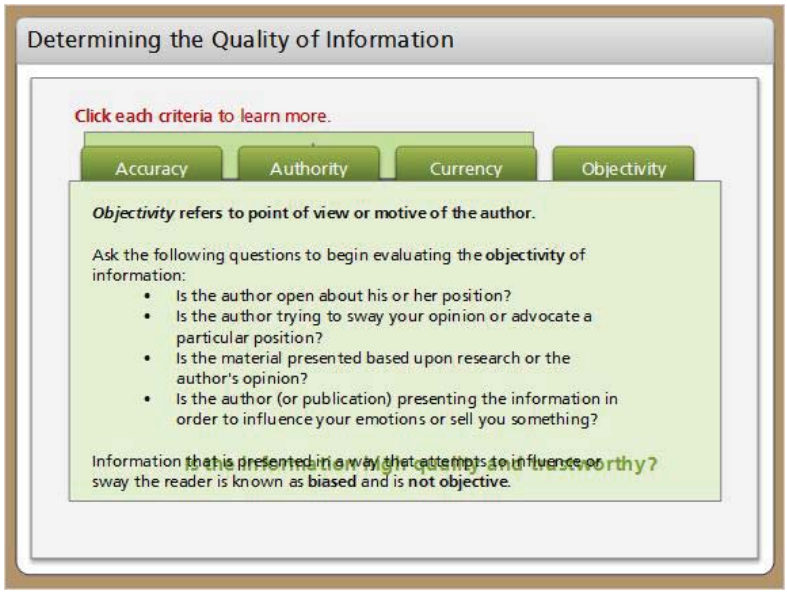
Authority (Slide Layer)



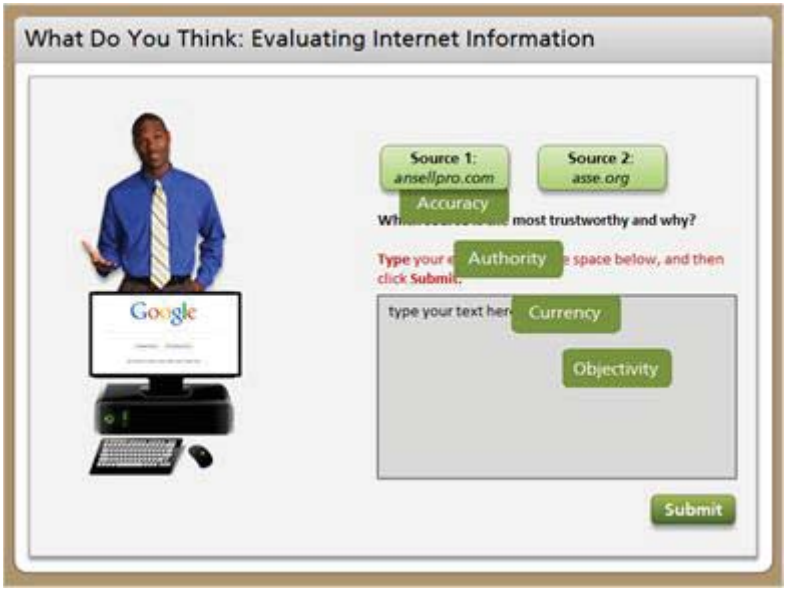
Currency (Slide Layer)



Objectivity (Slide Layer)



3.3 What Do You Think: Evaluating Internet Information



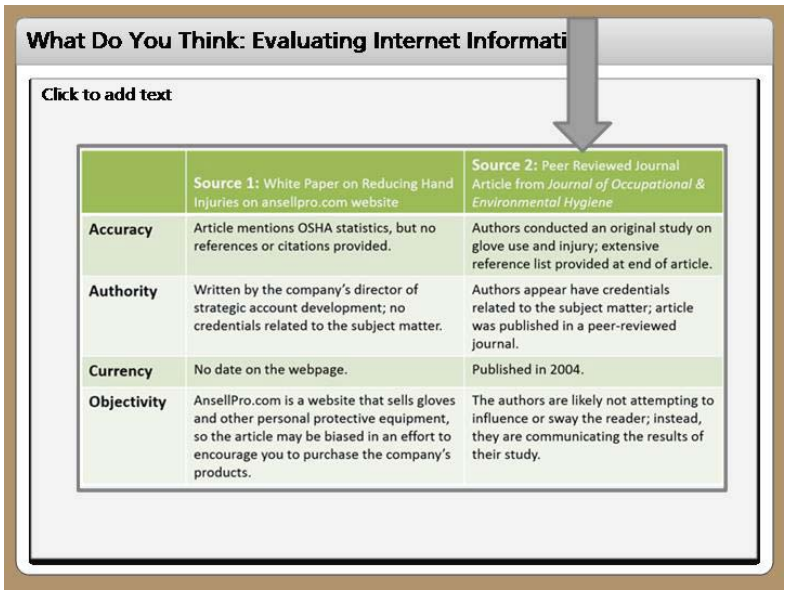
Notes:

The guidelines related to accuracy, authority, currency, and objectivity apply to both print sources and online sources, but there are a few additional guidelines you should consider when evaluating information found on the Internet. Let's look at an example.

Jeff is searching on the internet for information about hand injuries. Click the buttons to view two sources of information Jeff has found. In the space provided, type 1-2 sentences to explain which source is the most trustworthy and why. Then click submit.

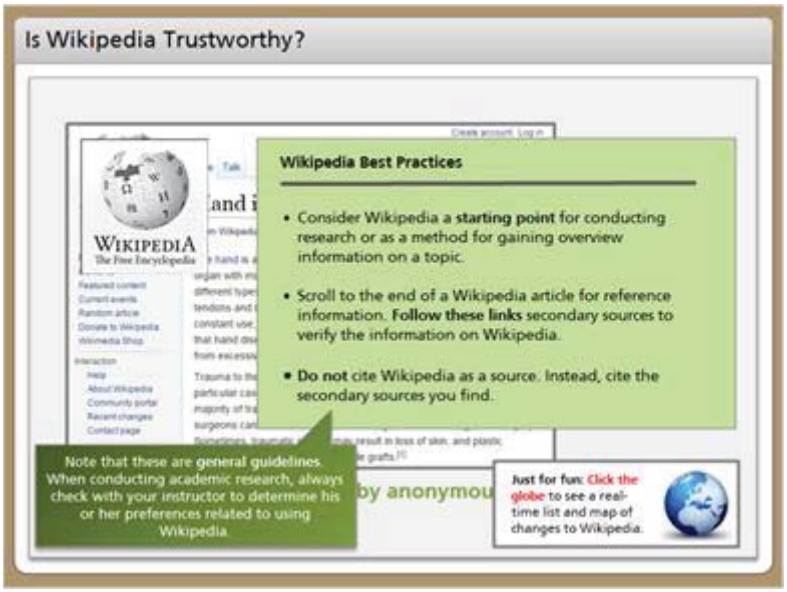
If you selected Source 2 as the most trustworthy, you are correct. Review the information shown here, which compares the accuracy, authority, currency, and objectivity of the two sources. Then click Next to continue.

Submit (Slide Layer)



	Source 1: White Paper on Reducing Hand Injuries on ansellpro.com website	Source 2: Peer Reviewed Journal Article from *Journal of Occupational & Environmental Hygiene*
Accuracy	Article mentions OSHA statistics, but no references or citations provided.	Authors conducted an original study on glove use and injury; extensive reference list provided at end of article.
Authority	Written by the company's director of strategic account development; no credentials related to the subject matter.	Authors appear have credentials related to the subject matter; article was published in a peer-reviewed journal.
Currency	No date on the webpage.	Published in 2004.
Objectivity	AnsellPro.com is a website that sells gloves and other personal protective equipment, so the article may be biased in an effort to encourage you to purchase the company's products.	The authors are likely not attempting to influence or sway the reader; instead, they are communicating the results of their study.

3.4 Is Wikipedia Trustworthy?



Notes:

People commonly use Wikipedia to find information. According to Wikipedia itself, Wikipedia is written collaboratively by anonymous volunteers. Currently, Wikipedia has more than 76,000 active contributors, and many Wikipedia entries are updated on an almost daily basis.

But is Wikipedia a trustworthy source of information? The best answers to this question are "maybe"

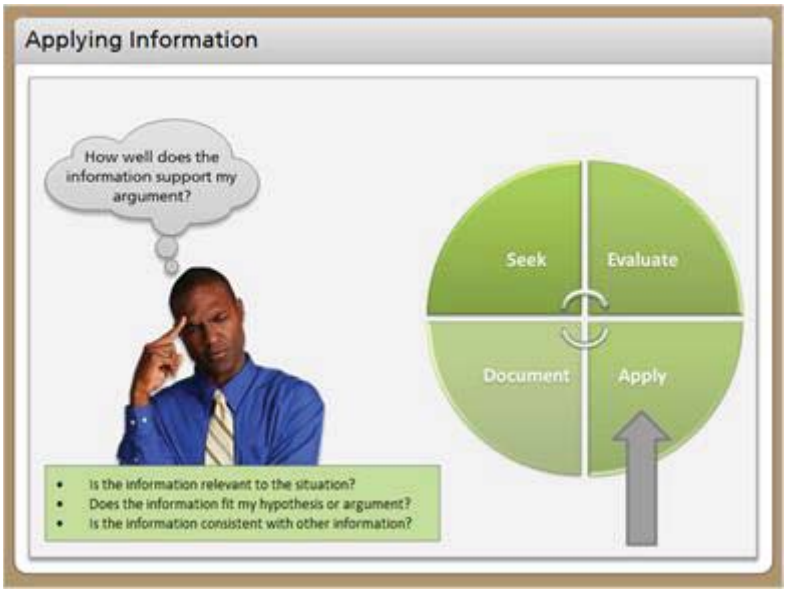
and “sometimes.”

Wikipedia can be very convenient if you are looking for basic reference information. But because Wikipedia is easily edited, it is generally not considered reliable, particularly for academic research.

Best practices to remember when using Wikipedia are shown here. Review these, and then click Next to continue.

4. APPLY

4.1 Applying Information

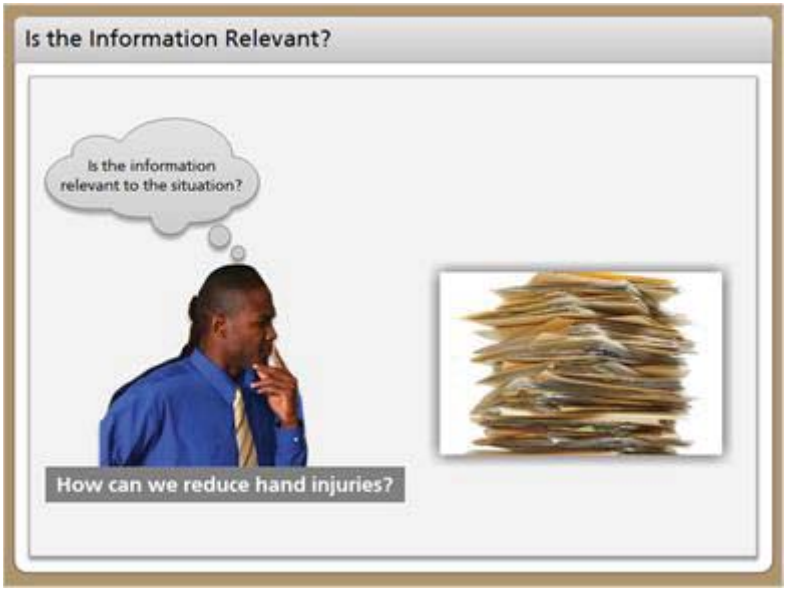


Notes:

Now that Jeff has found information on reducing hand injuries and evaluated this information, he needs to apply the information to the situation, determine if the information is relevant, and determine how well the information supports his argument. This is the third information literacy task.

Additional questions to ask related to this task are shown here.

4.2 Is the Information Relevant?



Notes:

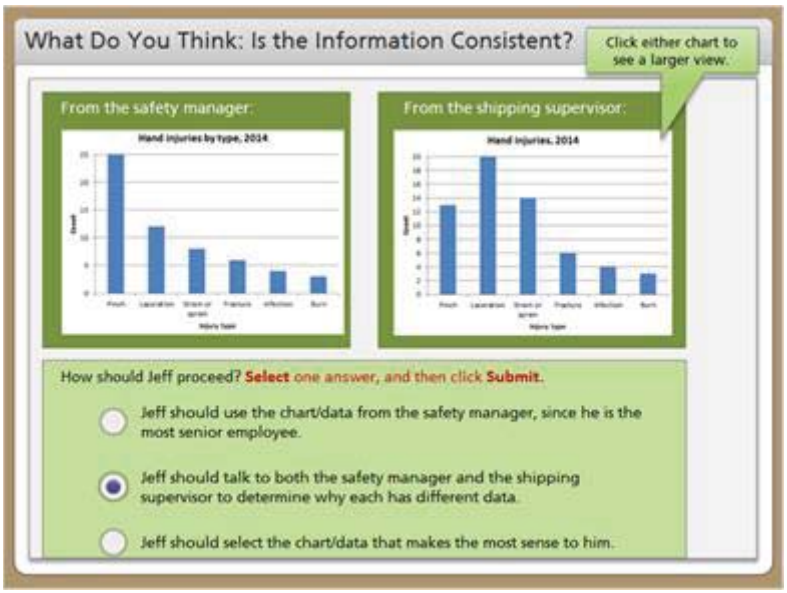
One of the questions to ask listed on the pervious page was "Is the information relevant to the situation?"

To be useful to you, the information you find must be central to your argument, not merely on the same general topic.

For example, imagine that the safety manager gives Jeff a stack of information and reports related to safety. While some of this information may be relevant to Jeff's task, Jeff will have to carefully review all information he is provided and finds on his own to ensure that it is relevant.

4.3 What Do You Think: Is the Information Consistent?

(Multiple Choice, 10 points, 1 attempt permitted)



Correct	Choice
	Jeff should use the chart/data from the safety manager, since he is the most senior employee.
X	Jeff should talk to both the safety manager and the shipping supervisor to determine why each has different data.
	Jeff should select the chart/data that makes the most sense to him.

Feedback when correct:

That's right! You selected the correct response. Continue to the next page to learn more.

Feedback when incorrect:

You did not select the correct response. Continue to the next page to learn more.

Notes:

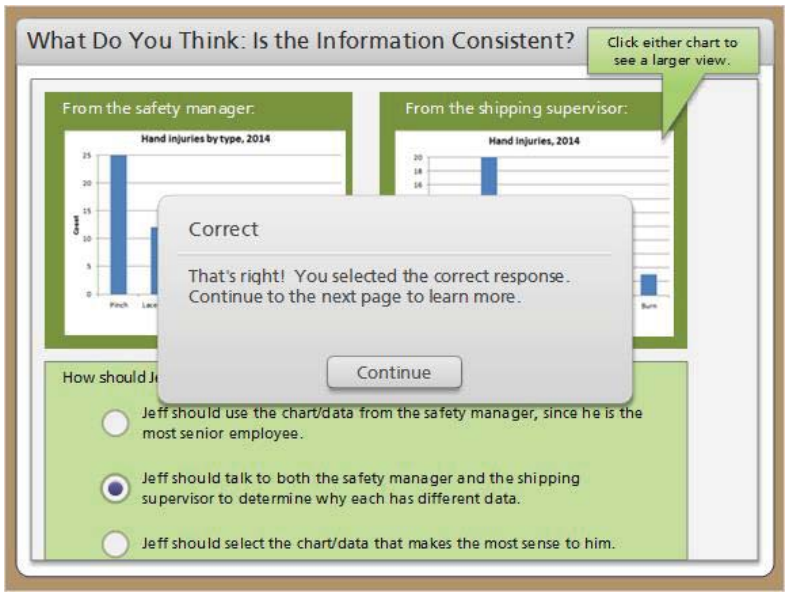
Let’s look at another example related to Jeff’s task.

Let’s say that Jeff has been provided this information by the current safety manager. However, the shipping supervisor provides him this information. Which of the option best describes how Jeff should

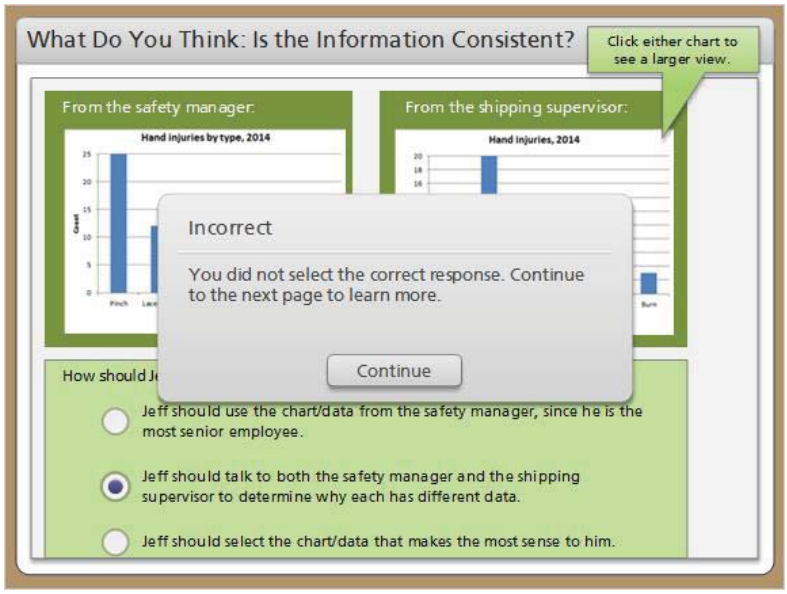
proceed?

Select one answer, and then click Submit.

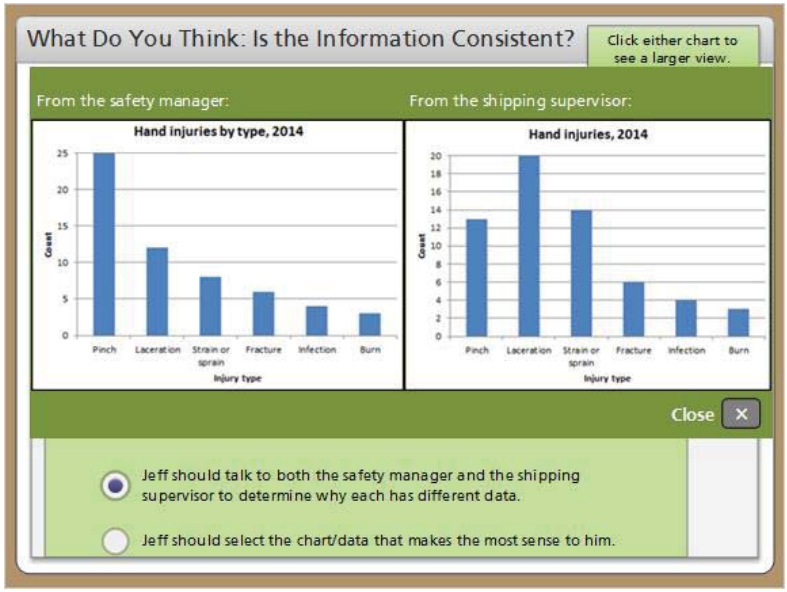
Correct (Slide Layer)



Incorrect (Slide Layer)



Larger Charts (Slide Layer)



4.4 Is the Information Consistent?

Notes:

If you said that Jeff should not proceed without talking to the safety manager and the shipping supervisor to clarify the discrepancies in the data, you are correct.

Review the guidelines here for applying information. Then click Next to continue.

5. DOCUMENT

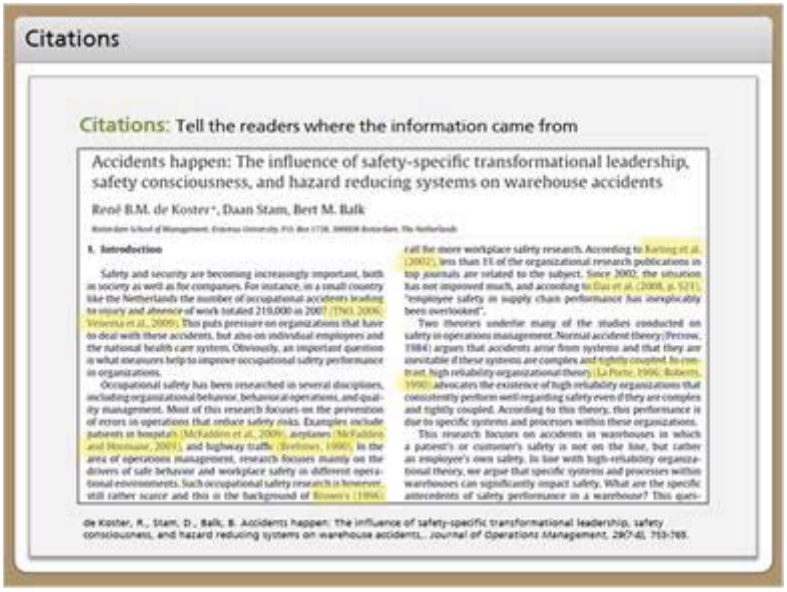
5.1 Documenting Information

Notes:

The last information literacy task is to document the information you have found.

This step is important for several reasons: it allows others to judge the quality of your information and build on your information. In addition, documenting your information acknowledges your sources of information.

5.2 Citations



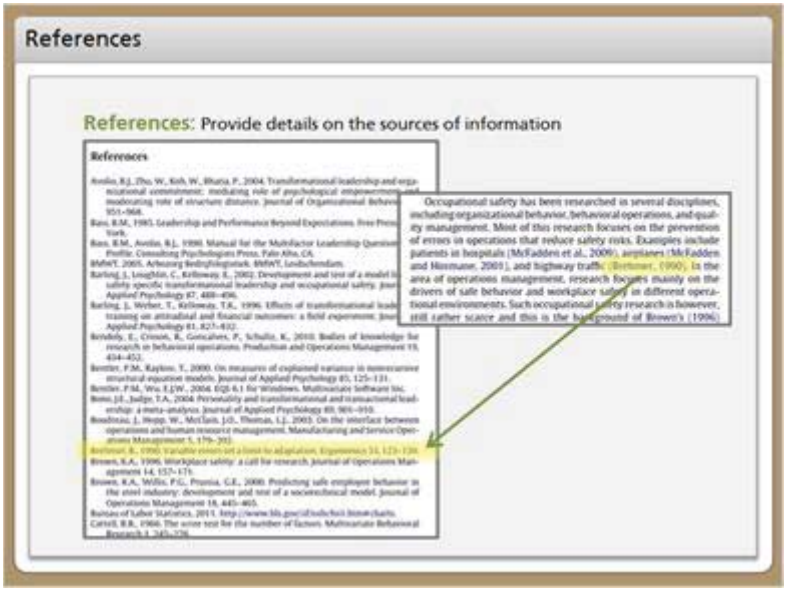
Notes:

In academic writing, you will typically document your sources uses citations and references.

Citations appear in the body of your writing. They tell the reader where the information came from.

The citations in this article on the topic of accidents in warehouses from the Journal of Operations Management are highlighted in yellow.

5.3 References

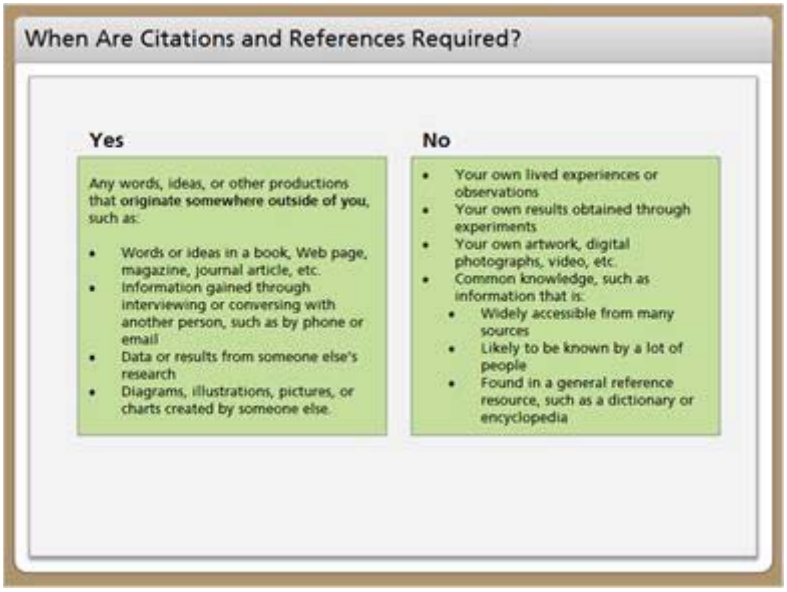


Notes:

References usually appear at the end of your writing. References provide details on the sources of information so readers could find the sources themselves if necessary. Shown here is the reference list for the Accidents Happen article you just saw.

For example, here is the reference list entry for Brehmer, 1990, which appears on the first page of the article.

5.4 When Are Citations and References Required?

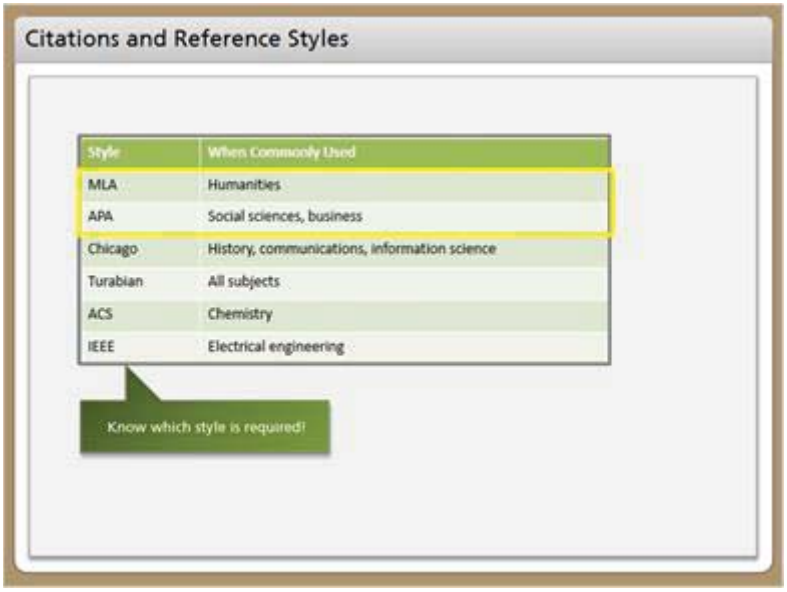


Notes:

A common question about citations and references is about when they are required. Guidelines for when citations and references are required in academic writing are shown here.

Review these guidelines, and then click Next to continue.

5.5 Citations and Reference Styles

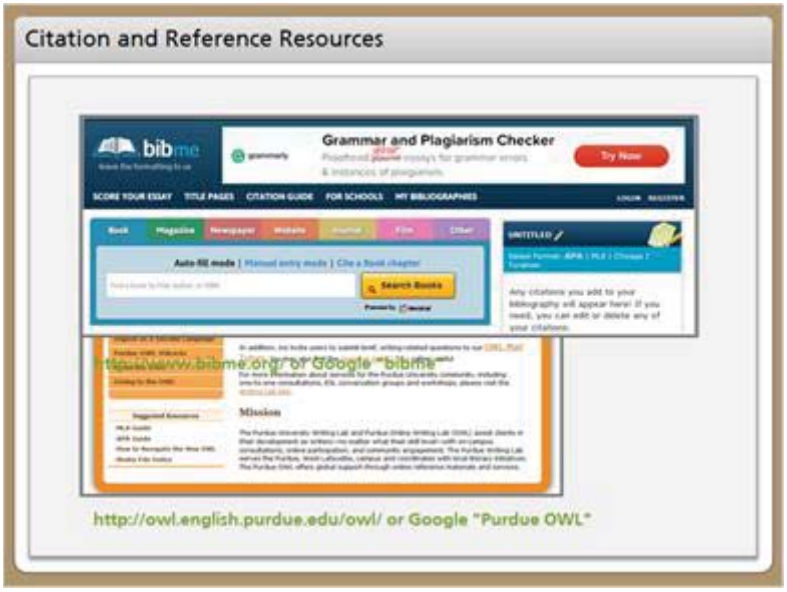


Notes:

In academic writing, multiple styles of citations and references exist; some are listed here. MLA and APA are the mostly commonly used styles.

A key point related to citation and reference styles is to know which style is required for each writing project. Your instructor will typically provide you this information.

5.6 Citation and Reference Resources



Notes:

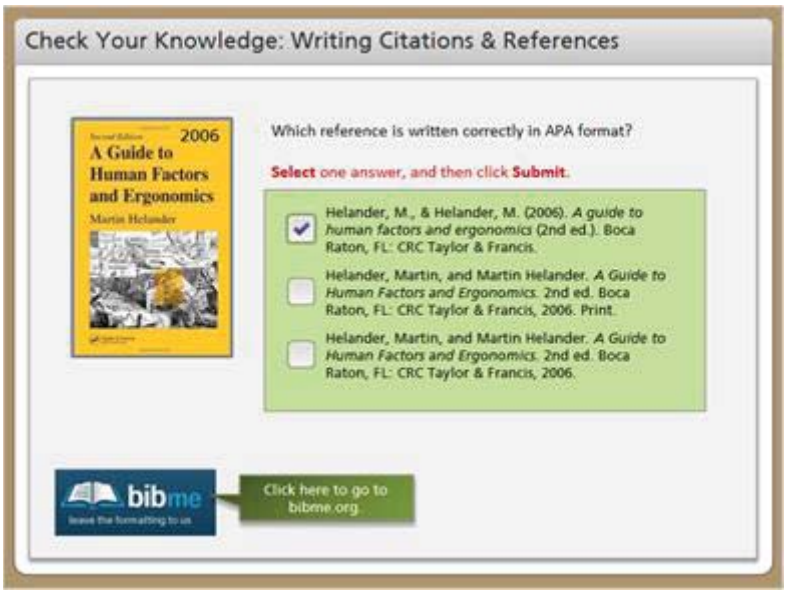
Many online resources that can help you write citations and references are available, so it isn't necessary for you to memorize citation and reference styles.

One resource that is widely used by college students at many universities is the Purdue Online Writing Lab, also known as the OWL. The OWL website contains extensive information on how to format citations and reference lists.

Another type of resource available to you is an online citation generation. When you use an online citation generator, you enter information about the source into a webpage, and the citation generator creates the reference citation for you. Bibme.org, the citation generator shown here, is a commonly used citation generator.

5.7 Check Your Knowledge: Writing Citations & References

(Multiple Response, 10 points, 1 attempt permitted)



Correct	Choice
X	Helander, M., & Helander, M. (2006). A guide to human factors and ergonomics (2nd ed.). Boca Raton, FL: CRC Taylor & Francis.
	Helander, Martin, and Martin Helander. A Guide to Human Factors and Ergonomics. 2nd ed. Boca Raton, FL: CRC Taylor & Francis, 2006. Print.
	Helander, Martin, and Martin Helander. A Guide to Human Factors and Ergonomics. 2nd ed. Boca Raton, FL: CRC Taylor & Francis, 2006.

Feedback when correct:

That's right! You selected the correct response. This citation is written correctly in APA format.

Feedback when incorrect:

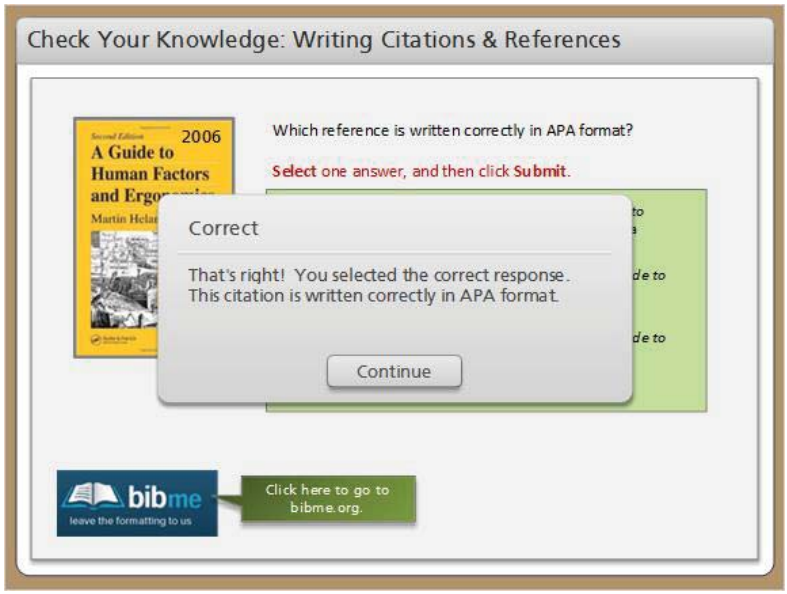
You did not select the correct response. The first citation is written correctly in APA format.

Notes:

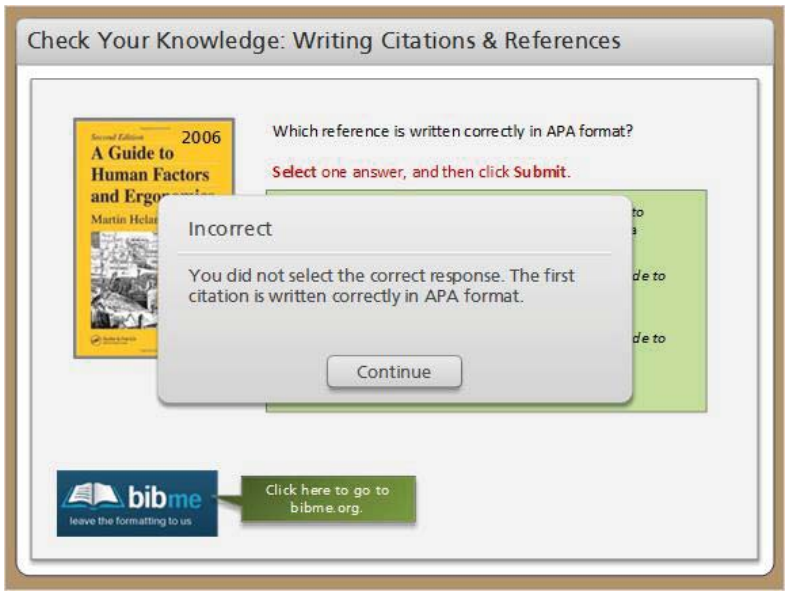
Imagine that Jeff has used some of the information in the book shown here to make evidence-based decisions related to his recommendations on reducing hand injuries.

Using bibme, determine which of the reference shown for this book is written correctly in APA format. Click **Submit** after making your choice.

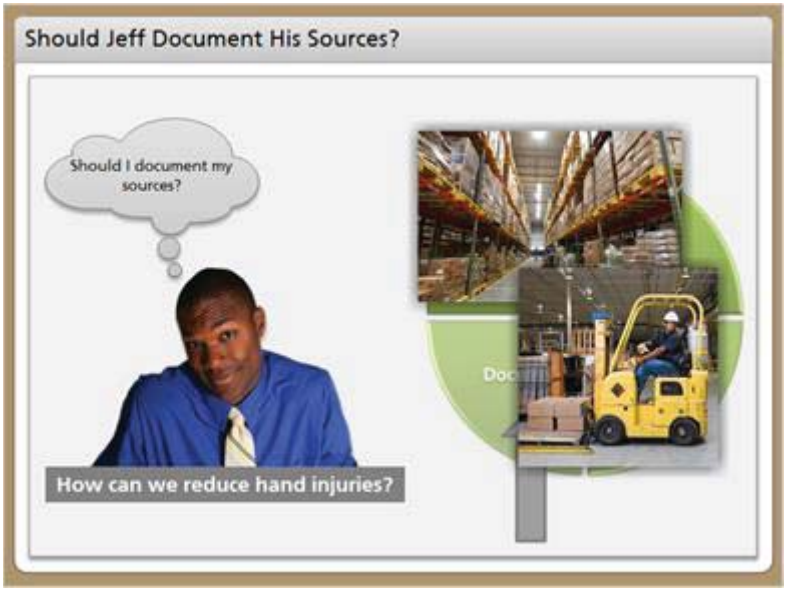
Correct (Slide Layer)



Incorrect (Slide Layer)



5.8 Should Jeff Document His Sources?



Notes:

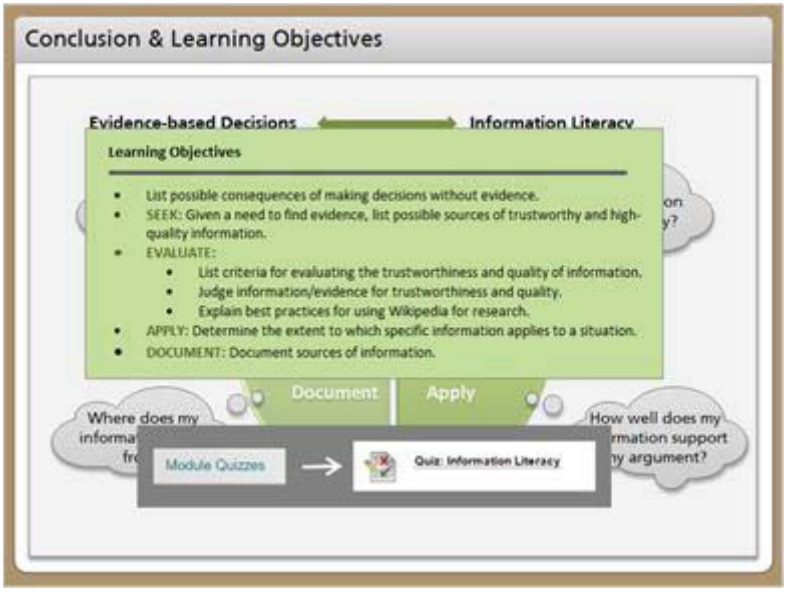
The more formal your writing task, the more important it is to cite your sources and include a reference list. However, if we think about Jeff's task to analyze the current situation related to hand injuries in the shipping and receiving areas and identify specific ways to reduce hand injuries, you might be wondering if Jeff needs to document his sources in his final report.

While Jeff should clarify expectations with the person who has assigned this task to him, it is probably a very good idea for him to document his sources. As you learned earlier, documenting sources allows others to judge the quality of the information and build on the information. In addition, documenting information acknowledges the sources of information.

For example, imagine that Jeff's final recommendations include modifying some of the equipment in the shipping and receiving area. The company will be likely be much more willing to implement recommendations that cost money if Jeff's recommendations are supported with high quality information and evidence.

6. Module Summary

6.1 Conclusion & Learning Objectives



Notes:

Congratulations. You've completed this online module! You should now have a clear understanding of the four information literacy tasks, their related questions, and how they relate to making evidence-based decisions.

To determine if you've mastered the learning objectives listed here, complete the Information Literacy quiz in your Blackboard course.

Remember that if you'd like to review any sections of this module, you can click the topics in the left-hand navigation.